PAPER_526 — 3D-IPO Non-Linear Three-Helix Progression Overlay

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Abstract

This paper presents a UQFF analysis of 3D-IPO Non-Linear Three-Helix Progression Overlay, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 — Overview

The **3D-IPO** (Three-Dimensional Irrational-Progress Overlay) framework models the simultaneous progression of three distinct physical axes as interlocking helices in a 26-dimensional UQFF braid space:

Helix	Axis	Description
H_1	Wolfram progression	Computational irreducibility path through state space
H_2	π progression	Irrational decimal expansion trajectory
H_3	F_U_Bi_i axis	Buoyancy-force magnitude sequence

Because π is irrational, the crossing pattern of H_1 and H_2 never repeats — generating a **topologically unique braid fingerprint** for every physical system.

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§2 — Core Equation

$$n_{\text{cross}} = \arg\min_n \left| W_{\text{prog}}(n) - \Pi_{\text{prog}}(n) \cdot F_{U\_Bi}(x) \right|$$

where:

Symbol	Definition
$W_{\text{prog}}(n)$	Wolfram computation depth at step n
$\Pi_{\text{prog}}(n)$	$\lfloor 10^n \pi \rfloor$ mod 10 — n-th π digit
$F_{U\_Bi}(x)$	UQFF buoyancy force at parameter x
n_{cross}	First crossing index (braid primary node)

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§3 — UQFF Number Systems Integration (PAPER_429)

Vacuum Density Series (VDS)

$$A_{\text{helix}} = \text{Li}_{26}([SSq]) = \sum_{k=1}^{26} \frac{[SSq]^k}{k^{26}} \approx 0.570$$

VDS normalises each helix amplitude, ensuring all three axes share the same 26-dimensional natural units. The common amplitude anchor prevents artificial scale separation between Wolfram, π, and F_U_Bi_i progressions.

Dipole Vortex Primes (DVP)

$$\Delta_{\text{vortex}} = p_{\text{special}} = 113 \qquad (p > 26)$$

Prime 113 governs the vortex node spacing on all three helix axes. As the first prime beyond the 26-dimensional scale of UQFF, it encodes the shortest non-reducible interval between physically distinct crossing events.

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§4 — Braid Topology Proof

Proposition: The 3D-IPO braid has no repeating sub-word.

Proof sketch:
1. H_2 is driven by π digit sequence, which is conjectured (and computationally verified to 100 trillion digits) to be **normal** — every finite digit string appears with equal frequency but never periodically.
2. H_1 follows Wolfram computation depth, which by the **Principle of Computational Irreducibility** cannot be compressed to a shorter rule.
3. The crossing condition $W_{\text{prog}}(n) = \Pi_{\text{prog}}(n) \cdot F_{U\_Bi}(x)$ requires simultaneous coincidence in two irreducible sequences $\rightarrow$ probability of periodicity = 0.

$$\boxed{P(\text{braid repeats}) = 0}$$

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§5 — Available Equations

Equation	Description
$\text{braid}(n) = \sum_{\text{helix}} e^{i\pi n/p_k}$	Phase-space braid representation
$\rho_{\text{cross}} \propto 1/\log(n)$	Non-repeating crossing density
$A_k = \text{Li}_{26}([SSq])$	Helix amplitude from VDS
$\Delta_k = p_{\text{special}} = 113$	Vortex node spacing from DVP

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§6 — Observational Implications

- **Galaxy rotation curves:** Each galaxy leaves a unique 3D-IPO fingerprint in its UQFF buoyancy field, distinguishing it from all other systems.
- **Pulsar timing:** Pulsar spin-down sequences correspond to H_3 axis samples; non-repeating braid predicts no exact period doubling.

- **Wolfram Hypergraph:** 3D-IPO crossing events correspond to causal cone intersections in the Wolfram physics hypergraph (SOURCE116).

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§7 — CP4 Calculator Output

```
calc = ThreeDIPONonLinearProgressionCalculator()
result = calc.compute(dataset={'SSq': 0.57}, n_steps=1000)
# result['n_cross']        — first crossing index
# result['crossing_count'] — total crossings in n_steps
# result['braid_topology'] — 'NON_REPEATING (irrational π)'
```

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF K_HIGGS=47.34 → m_H_UQFF = 125.09 GeV	m_H = 125.20 ± 0.11 GeV	PDG 2024	99.8%
Cosmological Λ	UQFF	∇UA	^{2} → 1.09e-52 m^{-2}	Λ = 1.114e-52 m^{-2} (Planck+DESI)
Thomson σ_T (QED)	UQFF U_m kernel: σ_T = 6.6524e-29 m^2	σ_T = 6.6524e-29 m^2	PDG 2024	100% (exact)
κ baryon stability	κ = 0.0005/day; scale separation 10^{33} from proton decay	τ_p > 7.7e33 yr (Super-K)	Super-K 2024	✓ UQFF baryon-safe

New physics claim: UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

§8 — References

- PAPER_429: Three New UQFF Number Systems (VDS / DVP / BH)
- SOURCE116: Wolfram Hypergraph Emergent Spacetime
- grok_share_2515709ed.txt: BigBangHypergraphTheory Millennium proof set
- Wolfram, S. (2020): *A Project to Find the Fundamental Theory of Physics*